test input, full runs — small-2way

L1D 32 sets x 2, L2 64 sets x 8 · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

35 of 35 runs completed, 0 running, 0 pending; **0 remaining**.

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	1/1	21.32%	51.31%	55.47%	21.42%	51.49%	55.80%
parest	1/1	0.22%	0.56%	1.65%	0.81%	1.48%	12.55%
deepsjeng	1/1	2.78%	9.33%	15.84%	7.31%	22.43%	46.66%
leela	1/1	0.36%	1.25%	1.66%	6.11%	15.91%	44.30%
namd	1/1	3.13%	7.91%	17.27%	3.14%	7.94%	17.37%
lbm	1/1	9.67%	27.79%	35.34%	9.67%	27.79%	35.34%
nab	1/1	7.90%	26.29%	39.86%	8.63%	27.88%	47.43%
xz	1/1	0.68%	2.35%	2.75%	3.30%	9.53%	17.13%
xz_4m_l1	1/1	0.59%	2.08%	2.43%	2.95%	8.51%	15.84%
xz_4m_l2	1/1	0.54%	1.91%	2.32%	2.73%	7.91%	15.62%
xz_4m_l3e	1/1	2.51%	9.02%	25.13%	3.76%	12.59%	42.10%
xz_4m_l4	1/1	2.12%	6.97%	14.43%	4.34%	13.23%	32.30%
xz_4m_l4e	1/1	2.28%	8.15%	25.17%	3.39%	11.30%	42.07%
xz_1m_l0	1/1	0.62%	2.50%	1.68%	3.69%	10.95%	17.49%
xz_1m_l1	1/1	0.58%	2.38%	1.48%	3.48%	10.36%	16.63%
xz_1m_l2	1/1	0.57%	2.35%	1.45%	3.39%	10.13%	16.34%
xz_1m_l3e	1/1	2.59%	9.35%	16.76%	4.60%	15.17%	33.91%
xz_1m_l4	1/1	2.01%	6.65%	9.30%	5.08%	15.11%	26.95%
xz_1m_l4e	1/1	2.43%	8.77%	16.84%	4.27%	14.07%	33.92%
imagick	1/1	0.02%	0.08%	0.05%	4.16%	11.90%	27.17%
blender	1/1	1.16%	2.72%	4.44%	9.37%	20.20%	39.50%
povray	1/1	0.27%	0.45%	0.93%	14.17%	22.21%	55.05%
omnetpp	1/1	0.09%	0.19%	0.36%	12.82%	28.51%	53.40%
xalancbmk	1/1	0.23%	0.53%	0.92%	8.96%	16.83%	45.24%
x264	1/1	6.24%	13.47%	39.69%	8.48%	18.06%	54.69%
gcc	1/1	0.84%	2.18%	3.58%	12.72%	34.38%	52.28%
mcf	1/1	1.58%	2.11%	16.47%	2.74%	3.87%	27.40%
perlbench_makerand	1/1	0.08%	0.00%	0.50%	18.63%	28.89%	59.00%
bwaves_1	1/1	3.55%	9.47%	18.87%	3.55%	9.47%	18.87%
bwaves_2	1/1	3.55%	9.47%	18.87%	3.55%	9.47%	18.87%
exchange2	1/1	5.03%	10.75%	12.58%	6.23%	13.14%	15.81%
fotonik3d	1/1	3.84%	10.81%	9.25%	3.87%	10.86%	9.40%
roms	1/1	4.79%	15.22%	9.67%	4.88%	15.37%	10.29%
wrf	1/1	0.91%	3.41%	2.49%	1.59%	4.94%	7.83%
cam4	1/1	2.11%	7.90%	7.04%	4.52%	14.20%	21.40%
pooled		5.17%	12.33%	27.44%	7.17%	16.47%	39.99%
mean		2.78%	7.88%	12.36%	6.35%	15.89%	31.03%

Miss rates

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.76%	4.45%	35.78%	3.91%	14.56%	64.32%	13.52%	8.96%
parest	done	11.59%	9.58%	12.36%	3.51%	2.72%	57.67%	0.55%	2.81%
deepsjeng	done	9.98%	2.86%	12.90%	7.61%	4.89%	6.89%	1.74%	13.49%
leela	done	6.90%	9.70%	7.74%	0.42%	4.44%	20.72%	0.00%	96.05%
namd	done	12.89%	5.49%	16.11%	1.31%	2.90%	25.71%	4.25%	17.28%
lbm	done	23.74%	0.42%	8.74%	2.26%	79.52%	88.75%	0.00%	3.10%
nab	done	6.91%	1.39%	10.46%	2.12%	0.91%	4.63%	0.13%	0.50%
xz	done	6.50%	0.12%	8.27%	1.00%	3.39%	25.90%	0.01%	4.24%
xz_4m_l1	done	8.00%	0.12%	9.43%	0.81%	5.52%	35.15%	0.01%	7.68%
xz_4m_l2	done	9.80%	0.12%	11.81%	0.81%	5.96%	42.07%	0.01%	8.16%
xz_4m_l3e	done	6.92%	1.91%	8.00%	1.17%	5.42%	19.19%	0.00%	1.06%
xz_4m_l4	done	8.37%	2.25%	10.00%	2.39%	5.73%	30.25%	0.00%	2.97%
xz_4m_l4e	done	6.70%	2.02%	7.58%	1.17%	5.51%	17.71%	0.00%	0.95%
xz_1m_l0	done	5.94%	0.17%	8.11%	2.01%	2.26%	29.24%	0.02%	2.93%
xz_1m_l1	done	6.73%	0.17%	8.81%	1.71%	3.19%	33.78%	0.02%	3.89%
xz_1m_l2	done	6.93%	0.15%	9.07%	1.72%	3.25%	34.23%	0.02%	3.97%
xz_1m_l3e	done	6.21%	1.54%	7.72%	2.13%	3.74%	18.66%	0.00%	0.46%
xz_1m_l4	done	6.83%	1.73%	8.77%	4.36%	3.70%	24.25%	0.00%	0.93%
xz_1m_l4e	done	6.17%	1.62%	7.52%	2.11%	3.79%	17.52%	0.00%	0.43%
imagick	done	4.42%	10.07%	3.82%	69.57%	5.76%	81.36%	35.66%	54.63%
blender	done	5.34%	2.28%	5.62%	3.16%	5.05%	19.43%	1.54%	1.25%
povray	done	10.14%	7.41%	12.18%	20.68%	4.98%	41.21%	67.06%	66.16%
omnetpp	done	12.56%	0.25%	16.96%	0.86%	4.40%	26.59%	20.83%	3.14%
xalancbmk	done	13.66%	10.00%	16.91%	18.70%	4.80%	33.74%	4.23%	4.16%
x264	done	2.33%	0.39%	2.80%	0.95%	2.40%	26.97%	1.46%	13.50%
gcc	done	7.17%	0.67%	9.36%	2.17%	4.21%	32.52%	5.35%	11.67%
mcf	done	25.50%	5.57%	28.54%	8.28%	9.83%	75.88%	6.15%	15.64%
perlbench_makerand	done	5.72%	0.00%	8.03%	2.72%	1.02%	0.00%	—	0.00%
bwaves_1	done	41.07%	2.70%	45.88%	0.33%	44.79%	31.22%	10.11%	0.24%
bwaves_2	done	41.07%	2.70%	45.89%	0.33%	44.79%	31.22%	10.11%	0.24%
exchange2	done	2.50%	2.74%	3.18%	3.56%	1.38%	1.08%	0.11%	0.44%
fotonik3d	done	17.86%	5.71%	24.31%	0.06%	6.13%	41.35%	1.22%	2.98%
roms	done	21.00%	4.44%	25.89%	1.56%	15.86%	38.82%	7.87%	9.26%
wrf	done	18.06%	9.40%	19.31%	6.61%	15.08%	22.28%	6.65%	10.05%
cam4	done	10.24%	2.66%	10.53%	2.91%	11.62%	52.99%	6.95%	31.72%
pooled		8.77%	2.14%	10.92%	1.96%	6.92%	38.69%	8.40%	10.40%
mean		11.76%	3.22%	13.95%	5.29%	9.81%	32.95%	6.05%	11.57%

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save

column	formula	aciklama
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.